Zamiul Alam

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SUMMARY

Data Scientist with 4+ years of experience designing and deploying machine learning solutions across tabular data, time series, NLP, and clinical health informatics. Proven ability to articulate complex technical findings to diverse audiences, collaborate cross-functionally with engineers and domain experts, and solve regression, classification, clustering, forecasting, and anomaly detection problems. Passionate about using data-driven approaches to improve value-based care (VBC) outcomes in healthcare settings.

EDUCATION

Washington University in St Louis **Aug 2023 - Aug 2025**
Master's in Physics (CGPA: 3.51/4.00)

TECHNICAL SKILLS

- Languages & Platforms: Python, SQL, R, GCP, Tableau, Mathematica, MATLAB, Jupyter, Linux, Git, Word/Excel
- Python Libraries: Pandas, NumPy, scikit-learn, Matplotlib, Seaborn, Folium, XGBoost, CATBoost, PyTorch, TensorFlow
- Machine Learning: Time Series Analysis, Regression, Inference, Classification, Clustering, Anomaly Detection, Forecasting, A/B Testing & Experimentation, Boosted Trees, Neural Networks, Natural Language Processing (NLP)
- Agentic AI: Hugging Face, GPT, Llama, LangChain, Azure, Large Language Model (LLM), Model Context Protocol (MCP)
- Soft Skills: Communication, Collaboration, Adaptability, Leadership, Punctuality, Critical thinking, Creativity

PROFESSIONAL EXPERIENCE

Data Science and AI Researcher **Sep 2025 - Present**
- Collaborating with UChicago and Stanford researchers to develop and deploy ML models (ClinicalBERT, LSTM, XGBoost) on health informatics data including ICD and CPT coded records for diagnosis and procedure classification.
- Collaborating with University of Illinois Chicago researchers applying LLMs for framing analysis; presenting technical findings to interdisciplinary audiences including the American Political Science Association (APSA).

Adjunct Faculty (Harper College) **Nov 2025 - Present**
- Designing coursework and teaching statistics, math, and physics to a diverse student body, including future health professionals in Sonography and Radiography programs.
- Communicate complex quantitative concepts clearly to non-technical students, reinforcing skills in translating data-driven work for diverse audiences.

Arts & Sciences Graduate Fellow (Washington University in St. Louis**)** **Aug 2023 - Aug 2025**
- Conducted research using statistical modeling, multivariate analysis, and numerical simulations; applied stochastic and regression models to large scientific datasets.
- Communicated highly technical research results to faculty, senior staff, and peers, translating complex findings into clear, actionable insights for non-specialist audiences.

Research Assistant (Northern Illinois University) **Aug 2021 - Aug 2023**
- Applied chi-squared tests, regression models, and statistical sampling to fit large particle physics datasets; optimized pipeline to reduce computation time by 90%.
- Demonstrated leadership and mentorship by teaching physics lab sections and guiding students in Python-based data analysis and visualization.

SELECTED DATA PROJECTS

ICU Mortality Prediction [Link]
- Cleaned and processed MIMIC-III (Medical Information Mart for Intensive Care) dataset with more than 2 million rows of hourly time series data of 38,597 distinct adult patients and 49,785 hospital admissions.
- Built and compared classification models (Logistic Regression, Random Forest, XGBoost, LSTM, ClinicalBERT) to predict ICU patient mortality, covering tabular and time-series ML use cases. Achieved AUC scores of 0.83.
- Applied findings within a health informatics context to demonstrate how ML-driven mortality prediction can reduce ICU burden, improve resource allocation, and ultimately support better patient outcomes.

Sepsis Prediction in Patients [Link]

- Cleaned Physionet dataset for early prediction of sepsis in ICU patients using high-resolution EHR data with 1.5 million rows.
- Developed XGBoost classification model for early sepsis detection achieving AUC of 0.95; applied correlation analysis and feature engineering to high-resolution EHR data.
- Early detection model designed to give clinicians a critical time advantage, potentially reducing sepsis mortality by enabling intervention before condition escalates.

Finding Fraudsters [Link], [Slides]

- Built fraud detection models (XGBoost, LightGBM, CATBoost) on IEEE credit card fraud dataset with 590,000+ transactions.
- Dealt with severe class imbalance (3.5% fraud rate) and tuned hyperparameters to achieve a 93% ROC AUC.
- Model deployable in real-time transaction screening to automatically flag suspicious activity, reducing financial losses and minimizing manual review workload for fraud analysts.

Sentiment Analysis and Topic Modelling [Link],

- Built a topic modeling pipeline applied to New York Times articles about Artificial Intelligence
- Used LLM models like BERT and BERTopic to discover and visualize latent themes across a corpus of 10,000 articles.
- Pipeline enables media organizations or policy teams to automatically track how public narratives around AI evolve over time.

Chicago Crimes Data Analysis [Link]

- Analyzed crime data from the Chicago Data Portal which contained 8 million rows and developed SARIMA time series models
- Created clear interactive maps with Seaborn and Folium generating actionable predictions for public safety authorities.

Predicting Employee Attrition [Link]

- Built supervised (Logistic Regression, Random Forest, XGBoost) and unsupervised (K-Means) models on IBM HR dataset.
- Used scaling, one-hot encoding and clustering to uncover key attrition drivers (income, satisfaction, tenure).
- Gives HR leaders an early warning system to identify employees at flight risk, enabling targeted retention efforts before costly turnover occurs.

CERTIFICATIONS

- Data or Specimens Only Research (CITI Program):
  Credentialed to access restricted clinical datasets and trained in IRB-aligned human subjects research ethics consistent with the American Medical Informatics Association (AMIA) standards for biomedical data stewardship.
- The Erdős Institute Data Science Boot Camp:
  Completed a 3 month intensive Data Science Boot Camp, developing end-to-end workflows including data cleaning, exploratory data analysis, predictive modeling, and results presentation through a team-based project.
- IBM Data Science Professional Certificate:
  Completed a 4 month specialization on Data Science and Machine Learning with hands-on experience in Python, SQL, data analysis, visualization, and model development through cloud-based projects and a capstone.
- Natural Language Processing Specialization:
  Completed a 3 month specialization on the fundamentals of NLPs and LLMs.
- Fundamentals of Agents (Hugging Face):
  Learnt how to build AI agents using state-of-the-art libraries such as smolagents, Llama, and LangGraph.

PUBLICATIONS

- **Standard model at 200 GeV** - published in Physical Review D (Impact Factor 5.3)
- **Dark Matter Boosted by Pb Collisions** - submitted to Physical Review D (Impact Factor 5.3)
- **AI agents for ICU mortality prediction** - submitting to ML4H (Machine Learning for Health Symposium)